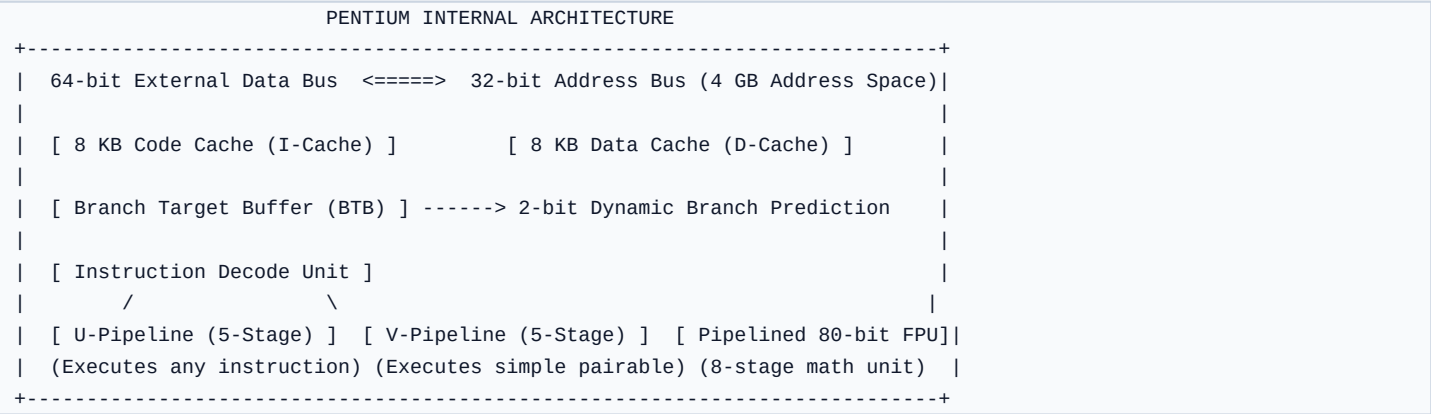


Pentium Processor Architecture & Superscalar Operation -- MPCA Module 5

Definition: The Intel Pentium Processor is a 32-bit superscalar CISC microprocessor capable of executing two instructions per clock cycle using two parallel 5-stage integer pipelines (U-Pipe and V-Pipe), an integrated on-chip Floating-Point Unit (FPU), and Dynamic Branch Prediction Logic.

1. Detailed Technical Explanation

1. Pentium Architecture Block Diagram



2. Superscalar Integer Pipelines (U and V Pipes)

The Pentium can issue two integer instructions simultaneously in parallel if they meet Instruction Pairing Rules:

1. U-

The 5 Pipeline Stages:

1. PF (Prefetch)	: Fetches instructions from 8 KB Code Cache into prefetch buffers.
2. D1 (Decode-1)	: Decodes instruction opcode, determines pairability between U & V.
3. D2 (Decode-2)	: Computes Effective Memory Address (Displacement + Base + Index).
4. EX (Execute)	: ALU executes integer operations; accesses Data Cache if needed.
5. WB (Writeback)	: Updates destination registers and status flags.

3. Dynamic Branch Prediction & Branch Target Buffer (BTB)

* Branch Target Buffer (BTB): An on-chip cache storing branch instructions, target jump addresses, and branch history bits.

* 2-b

4. Must-Write Points for Exams

* Pentium has a 64-bit external data bus and 32-bit address bus.

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5. Quick Recall Flow

Pentium -> Superscalar -> U & V Dual Pipelines (5-stages: PF, D1, D2, EX, WB) -> 8KB Code/Data Caches -> BTB 2-bit Branch Prediction